that leads to the formation of incomplete and defective polypeptide during translation.

**Comutagens**

The compounds which are not having own mutagenic properties but can enhance the effects of known mutagens are called comutagens.

Example: Ascorbic acid increase the damage caused by hydrogen peroxide.

Caffeine increase the toxicity of methotrexate

Mustard gas (Dichloro ethyl sulphide) used as chemical weapon in world war I.

H J Muller (1928) first time used X rays to induce mutations in fruit fly.

L J Stadler reported induced mutations in plants by using X rays and gamma rays.

Chemical mutagenesis was first reported by C. Auerback (1944).

### 3.5.3 Chromosomal mutations

The genome can also be modified on a larger scale by altering the chromosome structure or by changing the number of chromosomes in a cell. These large-scale variations are termed as **chromosomal mutations** or **chromosomal aberrations**. Gene mutations are changes that take place within a gene, whereas chromosomal mutations are changes to a chromosome region consisting of many genes. It can be detected by microscopic examination, genetic analysis, or both. In contrast, gene mutations are never detectable microscopically. Chromosomal mutations are divided into two groups: changes in chromosome number and changes in chromosome structure.

**I. Changes in chromosome number**

Each cell of living organisms possesses fixed number of chromosomes. It varies in different species. Even though some species of plants and animals are having identical number of chromosomes, they will not be similar in character. Hence the number of chromosomes will not differentiate the character of species from one another but the nature of hereditary material (gene) in chromosome that determines the character of species.

Sometimes the chromosome number of somatic cells are changed due to addition or elimination of individual chromosome or basic set of chromosomes. This condition in known as **numerical chromosomal aberration** or **ploidy**. There are two types of ploidy.

(i). Ploidy involving individual chromosomes within a diploid set (**Aneuploidy**)

(ii). Ploidy involving entire sets of chromosomes (**Euploidy**) (Figure 3.10)

**(i) Aneuploidy**

It is a condition in which diploid number is altered either by addition or deletion of one or more chromosomes. Organisms showing aneuploidy are known as **aneuploids** or **heteroploids**. They are of two types, Hyperploidy and Hypoploidy (Figure 3.11).

**1. Hyperploidy**

Addition of one or more chromosomes to diploid sets are called **hyperploidy**. Diploid set of chromosomes represented as Disomy. Hyperploidy can be divided into three types. They are as follows,

**(a) Trisomy**

Addition of single chromosome to diploid set is called **Simple trisomy**(2n+1). Trisomics were first reported by Blackeslee (1910) in ***Datura stramonium*** (Jimson weed). But later it was reported in ***Nicotiana, Pisum*** and ***Oenothera.*** Sometimes addition of two individual chromosome from different chromosomal pairs to normal diploid sets are called **Double trisomy** (2n+1+1).

**(b) Tetrasomy**

Addition of a pair or two individual pairs of chromosomes to diploid set is called **tetrasomy** (2n+2) and **Double tetrasomy** (2n+2+2)



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